GeoEnrich-NYC: A Geospatial Text-Tabular Dataset for Airbnb Price Prediction

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Abstract

We present GeoEnrich-NYC, a text-tabular dataset of 14,990 New York City Airbnb listings for nightly-price regression, in which each listing's coordinates are replaced by an LLM-generated description of its surroundings, built from Overture Maps places data. Evaluated with the MulTaBench curation pipeline, the dataset passes both curation criteria on all five learners: adding the description to the structured columns lowers RMSE on log price by 0.059–0.075, and fine-tuning the text encoder on the target lowers it by a further 0.002–0.007. The enrichment pipeline can be applied to other datasets with latitude-longitude coordinates, though we evaluate it on a single city.

1 Introduction

Text-tabular benchmarks need datasets in which the text column carries signal that the structured columns lack. MulTaBench [Arazi et al., 2026] turns this into a curation criterion: a dataset is admitted only if, across a committee of tabular learners, adding frozen text embeddings to the structured columns beats both unimodal baselines, and target-aware representations (TAR), text embeddings fine-tuned on the target, beat frozen ones. Neither Airbnb dataset MulTaBench evaluated passed, even though both include free-text listing fields.

Accommodation listings do carry a signal that their text fields rarely describe: location, a known determinant of nightly price [Wang and Nicolau, 2017]. A listing in downtown Manhattan, surrounded by restaurants and tourist attractions, should command a higher price than one in an outer neighbourhood with few amenities nearby. Yet datasets scraped from platforms such as Airbnb and Booking.com record location only as latitude-longitude coordinates, which say nothing directly about what surrounds a listing. We ask whether a natural-language description of those surroundings can supply this signal, that is, whether adding it to the structured columns lowers held-out error in nightly-price regression.

We build GeoEnrich-NYC to test this. Starting from Inside Airbnb's New York City listings [Inside Airbnb, 2026], we join each listing to the Overture Maps places [Overture Maps Foundation, 2026] within 450 meters, reduce them to citywide percentile ranks over 20 place categories plus the nearest of 77 curated landmarks, phrase these as word bands, and prompt an LLM to describe the surroundings (Figure 1). We use text rather than adding the counts as numeric columns because the dataset targets text-tabular learning, and the word bands keep only what stands out against the rest of the city. Two design choices keep the evaluation clean: the LLM never sees coordinates, exact counts, exact distances, or price, so the description cannot copy the target directly; and the coordinate columns are dropped, so location reaches the models only through the text (whether the text adds to coordinates is left to future work).

Our contributions are:

- **GeoEnrich-NYC**, a text-tabular regression dataset of 14,990 listings with 8 structured columns, one generated text column, and log nightly price as the target, released under CC BY 4.0 in MulTaBench's format.

- **A geospatial enrichment pipeline** that needs only latitude-longitude coordinates and withholds coordinates, exact measurements, and price from the LLM.
- **A MulTaBench evaluation** in which the dataset passes both curation criteria on all five learners (Section 4; limits in Section 5).

2 Related work

Text-tabular curation. We follow the MulTaBench protocol [Arazi et al., 2026], whose TAR condition builds on TabSTAR [Arazi et al., 2025], with the one adaptation described in Section 4. MulTaBench's curation rejected both Airbnb datasets it evaluated (Melbourne Airbnb, 1/5 learners; Airbnb Seattle, 0/5).

Location and LLMs in prediction. Hedonic pricing treats a property's price as the sum of implicit prices of its attributes, neighbourhood amenities among them [Rosen, 1974]. TabLLM [Hegselmann et al., 2023] serializes tabular rows into natural-language text and uses an LLM to predict from it; GeoLLM [Manvi et al., 2024] prompts an LLM with coordinates and nearby OpenStreetMap places to predict location-level indicators, including home values, drawing on the LLM's own geographic knowledge.

3 Dataset construction

3.1 Dataset and sources

We use Inside Airbnb's New York City listings snapshot [Inside Airbnb, 2026], scraped 14–23 June 2026 (CC BY 4.0), and query, with DuckDB, the places theme of Overture Maps release `2026-08-19.0` [Overture Maps Foundation, 2026] (CDLA-Permissive-2.0) for businesses and points of interest in the city.

3.2 Pipeline

The pipeline has four stages: data cleaning, geospatial enrichment, aggregation, and an LLM descriptive summary.

Data cleaning. We drop columns that do not partake in the regression task and enforce a schema. Listings with missing or non-positive prices are dropped, as are listings missing coordinates or more than 25% of their features. Of the 30,259 listings in the snapshot, 21,155 survive these filters (Table 3); we randomly sample 15,000 of them (seed 0), each with 8 feature columns (six numerical, one boolean, and one categorical) besides the target. This size matches MulTaBench's per-split cap of 10,000 training and 5,000 test rows, so the benchmark uses every row. The target is the log-transformed nightly price, $\log(1 + \text{price})$. Table 2 (Appendix A) summarizes the schema and feature distributions; the same appendix describes column parsing and missing-value handling.

Geospatial enrichment. We query Overture for all places within the listings' bounding box (padded by ≈ 2 km) that have a confidence score of at least 0.6 and belong to one of our selected categories. We apply a join such that each listing coordinate is assigned all POIs (points of interest) in a 450-meter radius (roughly 570 meters of walking on the Manhattan street grid), with distances measured in a metric projection (UTM zone 18N). The 10 listings with no POI within that radius are dropped, leaving 14,990 listings.

Aggregation. Each listing's nearby POIs are reduced to a count and the distance to the nearest place in each of 20 predefined categories (for example: restaurants, parks, ground transport, and nightlife venues). Those categories are curated from the level-2 categories of the official Overture hierarchical taxonomy. We rank each listing's count in every category, and its total count over all categories, against the citywide distribution of that count over all listings, and replace each count with its percentile. The ranking uses no price information, but is computed over all listings, including those later used for testing. Landmarks—77 curated attractions such as the Empire State Building, the Statue of Liberty, and Times Square—may carry stronger neighbourhood signal, so they are kept by name rather than reduced to categories, and matched over a wider 800-meter radius.

(1) Measured, with citywide rank	(2) Evidence line sent to the LLM
259 places within 450 m, 60th percentile	`Overall: busy, by how many places...`
4 discount stores, 97th percentile	`discount stores: far more than most neighbourhoods`
nearest historic site at 53 m ($\leq$ 150 m)	`historic sites: steps away`
Central Park at 792 m ($>$ 500 m)	`about 10 minutes from: Central Park`
e.g. 83 restaurants, 59th percentile	*16 categories within the typical range: omitted*

(3) Generated `surroundings_summary`: "This busy neighbourhood features far more discount stores than most neighbourhoods, with historic sites steps away. Central Park is about 10 minutes away."

Figure 1: How listing 10433 of the released dataset (entire home, 7 guests) gets its description. Each row traces one piece of evidence; colours mark where it surfaces in the summary. Of the 18 place categories near this listing, only one is a citywide contrast (outer 15%); typical ones are omitted.

LLM descriptive summary. Each aggregated record is rendered as a short *evidence block*, the only listing-specific input the LLM receives, in which word bands replace the numbers (Figure 1, column 2): an overall busyness band ("very quiet" to "very busy"), up to four category contrasts (citywide top or bottom 15%, or a common category absent nearby), at most one proximity cue within 150 meters ("steps away"), limited to telling place types such as parks and transit, and up to three landmarks within 800 meters in walking-distance bands ("right by" to "about 10 minutes from"). Typical categories are omitted, and the sections are shuffled per listing (seeded) so the summaries do not inherit the block's order (exact rules in Appendix B). The prompt asks for one or two neutral sentences that use only the given evidence and keep each comparison's strength. We chose Gemini 3.5 Flash-Lite [Google DeepMind, 2026] (`google/gemini-3.5-flash-lite`) over other LLMs tested on subsets for its relative lack of hallucinations, speed, and low cost, and fixed the prompt after testing 16 variants on $\approx$ 2,000 rows. Each listing is summarized independently through OpenRouter's batch API at temperature 0.4; generation is sampled, so reruns differ and the released text column is the reference.

The released dataset thus pairs the structured columns with a single text column, `surroundings_summary` (statistics in Appendix D). The code, prompts, word-band rules, and landmark list are available at `https://github.com/razilai/GeoEnrich`. The dataset is available on Hugging Face ([link]) in MulTaBench's format (typed `data.parquet` and `metadata.json`) under CC BY 4.0, attributing Inside Airbnb and the Overture Maps Foundation.

4 Results

With the prompt fixed, we evaluate GeoEnrich-NYC once with the MulTaBench benchmark, run from its current repository with no change to its evaluation code.[1] The curation committee consists of five learners: two GBDTs, LightGBM [Ke et al., 2017] and CatBoost [Prokhorenkova et al., 2018], the MLP ensemble TabM [Gorishniy et al., 2025], and the tabular foundation models TabPFNv2 [Hollmann et al., 2025] and TabPFNv2.5 [Grinsztajn et al., 2025]. Each is trained under four conditions: structured columns only, the surroundings description only (text-only), structured columns with frozen E5-small [Wang et al., 2022] text embeddings (joint frozen), and structured columns with E5-small embeddings fine-tuned on the target with low-rank adaptation [LoRA; Hu et al., 2022] (joint TAR). Every condition runs on the benchmark's six splits of TabArena's [Erickson et al., 2025] 67/33 scheme ($\approx$ 9,993 training and $\approx$ 4,997 test rows per split). The metric is the RMSE on the log-price scale, $\log(1 + \text{price})$, as reported by the current repository (the MulTaBench paper uses R^2); lower is better.

A learner shows *joint signal* when its mean joint-frozen RMSE is lower than both its structured and its text-only RMSE by more than 0.001, and *TAR gain* when its mean joint-TAR RMSE is lower than its joint-frozen RMSE by more than 0.001 (means rounded to three decimals before differencing). The dataset is eligible when at least 3 of the 5 learners show both. We adapt the acceptance criterion of Arazi et al. [2026, App. A.3] to RMSE rather than R^2; since the current repository no longer includes it, we compute it with our own code.

[1] `master` at commit `d88821d`; the dataset was added to its registry and served from a local Hugging Face cache.

Table 1: RMSE on $\log(1 + \text{price})$, mean $\pm$ standard deviation over six splits (lower is better). Δ_{joint}: improvement of joint frozen over the better unimodal baseline. Δ_{TAR}: improvement of joint TAR over joint frozen.

Learner	Structured	Text-only	Joint frozen	Joint TAR	Δ_{joint}	Δ_{TAR}
LightGBM	0.501 ± 0.007	0.726 ± 0.006	0.442 ± 0.006	0.435 ± 0.005	+0.059	+0.007
CatBoost	0.498 ± 0.006	0.722 ± 0.008	0.435 ± 0.005	0.431 ± 0.007	+0.063	+0.004
TabM	0.505 ± 0.008	0.720 ± 0.009	0.434 ± 0.004	0.432 ± 0.005	+0.071	+0.002
TabPFNv2	0.490 ± 0.008	0.715 ± 0.007	0.422 ± 0.005	0.418 ± 0.006	+0.068	+0.004
TabPFNv2.5	0.490 ± 0.007	0.710 ± 0.008	0.415 ± 0.004	0.412 ± 0.005	+0.075	+0.003

Table 1 reports the results. Adding the surroundings description to the structured columns lowers RMSE for every learner, by 0.059–0.075 (11.8–15.3% relative to the structured baseline), so all five learners show joint signal. The description alone is a weak predictor: text-only RMSE (0.710–0.726) is well above the structured baseline for every learner, which suggests the text complements the listing attributes, supplying the location information they lack by design, rather than substituting for them. Fine-tuning the encoder on the target lowers RMSE further for all five learners, so all five show TAR gain and the dataset passes the MulTaBench curation criterion with 5/5 learners on both checks. The TAR gain is small, however (0.002–0.007), and of the same order as the split-to-split standard deviation; joint TAR beats joint frozen on 27 of the 30 learner–split pairs (per-learner paired differences in Appendix C). The best configuration, TabPFNv2.5 with joint TAR, reaches an RMSE of 0.412, a 15.9% reduction over its structured baseline.

5 Limitations and future work

Evaluation. It is unclear what the main driver of the price signal is in each listing's surroundings description. It is possible that it is derived from the explicit naming of popular landmarks (present in the evidence of 42.6% of listings), or perhaps by the overall busyness band and the category contrasts. The pipeline already builds two censored arms that separate these sources — one keeping only the landmark names found in each summary, and one replacing each landmark name with "a well-known landmark" — but they were not evaluated in this release. The prompt was selected on rows that are also part of the released dataset, and was not re-validated on held-out rows. MulTaBench's random splits also place nearby listings, which share near-identical surroundings, in both training and test sets (12,594 distinct rendered evidence blocks for 14,990 listings), which may inflate the gains relative to unseen areas. Finally, Overture's places coverage is likely uneven across the city, so a "quiet" band may partly reflect sparse data.

Application. The pipeline needs only latitude-longitude coordinates, so it can be applied to other datasets, including Inside Airbnb's other cities, but we have evaluated it on a single city. The prompts are worded for hospitality and real-estate tasks, and the manually curated landmark list must be rebuilt for each city.

Future work. Future work could evaluate the pipeline beyond hospitality price regression. Since coordinates alone could carry the same or a stronger location signal, it could also rerun the four conditions with coordinates kept as structured columns and test whether TAR still beats joint frozen.

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A Data card

The bedroom and bathroom counts (`num_bedrooms`, `bathrooms`) are taken from the raw `bedrooms` and `bathrooms` columns, with missing bathroom counts parsed from the semi-structured `bathrooms_text` field (e.g., "1.5 baths") by a simple regex. Remaining missing values are kept as nulls: LightGBM, CatBoost, and TabPFN (Section 4) handle them natively, and the benchmark fills numerical nulls with training-set medians for TabM.

Table 2: Schema and feature distributions of the released dataset (14,990 rows). Quantiles are over non-missing values. The target is $\log(1 + \text{price})$, with price the nightly price in USD (raw quantiles: 7.2, 96, 173, 299, 30,973).

Column	Type	Missing	Min	Q1	Median	Q3	Max
`price` (target)	log USD	0.0%	2.10	4.58	5.16	5.70	10.34
`ratings`	numeric	27.8%	1.00	4.67	4.86	5.00	5.00
`guests`	numeric	0.0%	1	2	2	4	16
`bathrooms`	numeric	3.6%	0	1	1	1	15.5
`num_bedrooms`	numeric	33.8%	0	1	1	2	19
`property_number_of_reviews`	numeric	0.0%	0	0	6	41	4502
`minimum_nights`	numeric	0.0%	1	30	30	30	365
`room_type`	categorical	0.0%	Entire home 54.7%, Private 42.1%, Hotel 2.3%, Shared 0.9%				
`is_superhost`	boolean	0.0%	True 31.1%				
`surroundings_summary`	text	0.0%	14,254 distinct				

Table 3: Row attrition from the Inside Airbnb snapshot to the released dataset, and duplicates.

Step	Rows
NYC snapshot (scraped 14–23 June 2026)	30,259
Positive, non-missing price	21,515
Coordinates present and at most 25% of features missing	21,155
Random sample (seed 0)	15,000
At least one POI within 450 m (released)	14,990
Rows whose 8 tabular features and summary duplicate another row	46

B Evidence-block rules

The evidence block has up to four sections, each built from the aggregated record with word bands in place of numbers: (i) an overall line gives the citywide quintile of the total place count, from "very quiet" through "quiet", "average", and "busy" to "very busy"; (ii) a category is listed only if its percentile lies in the outer 15% (at or above the 85th, with at least three places, or at or below the 15th), as "more than most neighbourhoods" ("far more than most neighbourhoods" from the 95th, "more than almost anywhere in NYC" from the 99th), "fewer than most neighbourhoods" ("very few" up to the 5th), or "none nearby" for a category that at least 60% of listings have nearby; typical categories are omitted, and at most the four strongest contrasts are kept, one per level-1 category; (iii) at most one proximity cue marks the nearest park, museum, hotel, nightlife venue, historic site, or ground-transport service (rail, transit, or taxi) within 150 meters as "on the doorstep" (up to 50 meters) or "steps away", skipping categories already listed in (ii); (iv) up to three nearest landmarks within 800 meters are named (otherwise the block states that none is nearby), with their distance phrased as "right by", "a couple minutes from", "a short walk from", or "about 10 minutes from" (up to 100, 250, 500, and 800 meters, respectively). These sections, and the contrasts in (ii), appear in a per-listing shuffled order (seeded, so reproducible), so the summaries do not inherit a fixed evidence structure.

C Paired TAR gains

Table 4: Per-learner paired difference between joint frozen and joint TAR RMSE on $\log(1 + \text{price})$, computed split by split over the six splits (positive means TAR is better).

Learner	Paired difference, mean ± SD	Splits won by TAR
LightGBM	0.0074 ± 0.0024	6/6
CatBoost	0.0047 ± 0.0032	5/6
TabM	0.0014 ± 0.0033	4/6
TabPFNv2	0.0035 ± 0.0017	6/6
TabPFNv2.5	0.0036 ± 0.0010	6/6

D Summary text statistics

Table 5: Statistics of `surroundings_summary` over the 14,990 released rows. Words are alphabetic tokens. Jaccard similarity is computed on content-word sets (stopwords and generic words such as "area" and "neighbourhood" removed) over 20,000 random pairs (seed 0); lower means more varied text.

Statistic	Value
Words per summary, mean ± SD (median; min–max)	39.1 ± 11.1 (38; 11–84)
Characters per summary, mean (median; max)	238.7 (229; 522)
Sentences per summary: 1 / 2 / 3	450 / 14,071 / 469
Distinct summaries	14,254 (95.1%)
Distinct rendered evidence blocks	12,594 (84.0%)
Vocabulary (lower-cased word types)	442
Mean pairwise content-word Jaccard similarity	0.185
Listings with a landmark in the evidence	42.6%
of which the summary names at least one landmark	99.6%
Summaries containing "quiet" / "busy"	39.4% / 39.9%